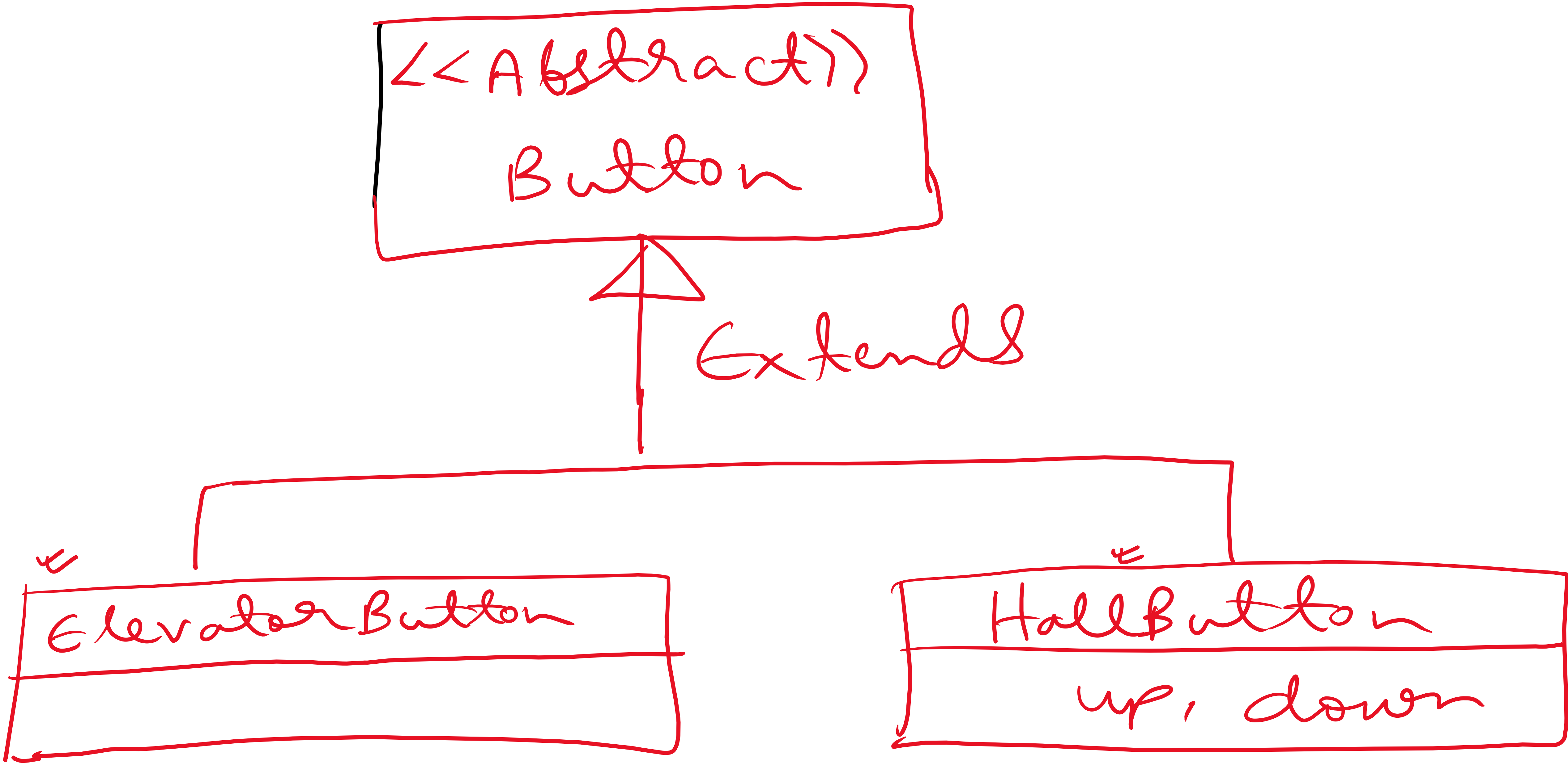

<<Abstract>>
Button
Extends
ElevatorButton
HallButton
up, down

Elevator Panel

floorButton
openButton
closeButton

HallPanel

up
down

Display

floor : int
capacity : int
direction : Direction

ShowElevatorDisplay()
ShowHallDisplay()

Door
state : DoorState
isOpen()

ElevatorCar
id, door : Door, state : ElevatorState, display : Display, panel : EP, currentFloor : int
openDoor(), closeDoor(), move(), stop().

Floor

displays : Display (list)
panel : HallPanel (list)

isBottomMost() : bool
isTopMost() : bool

Building

floors : Floor (list)
elevators : ElevatorCar (list)

Elevator System

building : Building

monitoring()

selectBestElevatorCar()

<<enumeration>>
ElevatorState

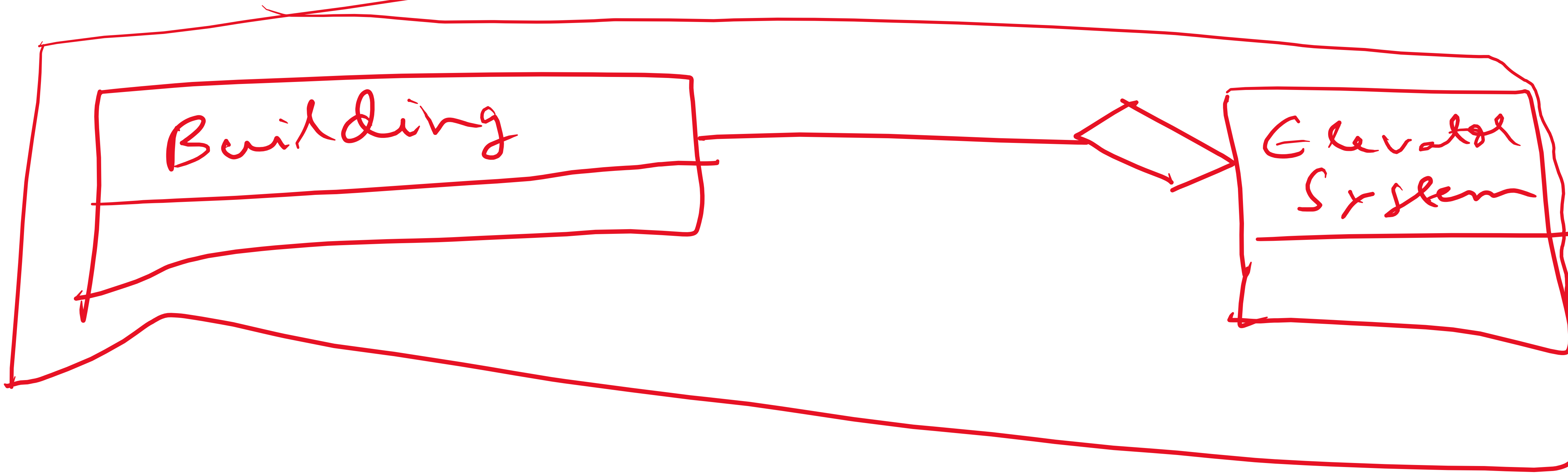
idle, up, down

<<enumeration>>
Direction

up, down

<<enumeration>>
DoorState

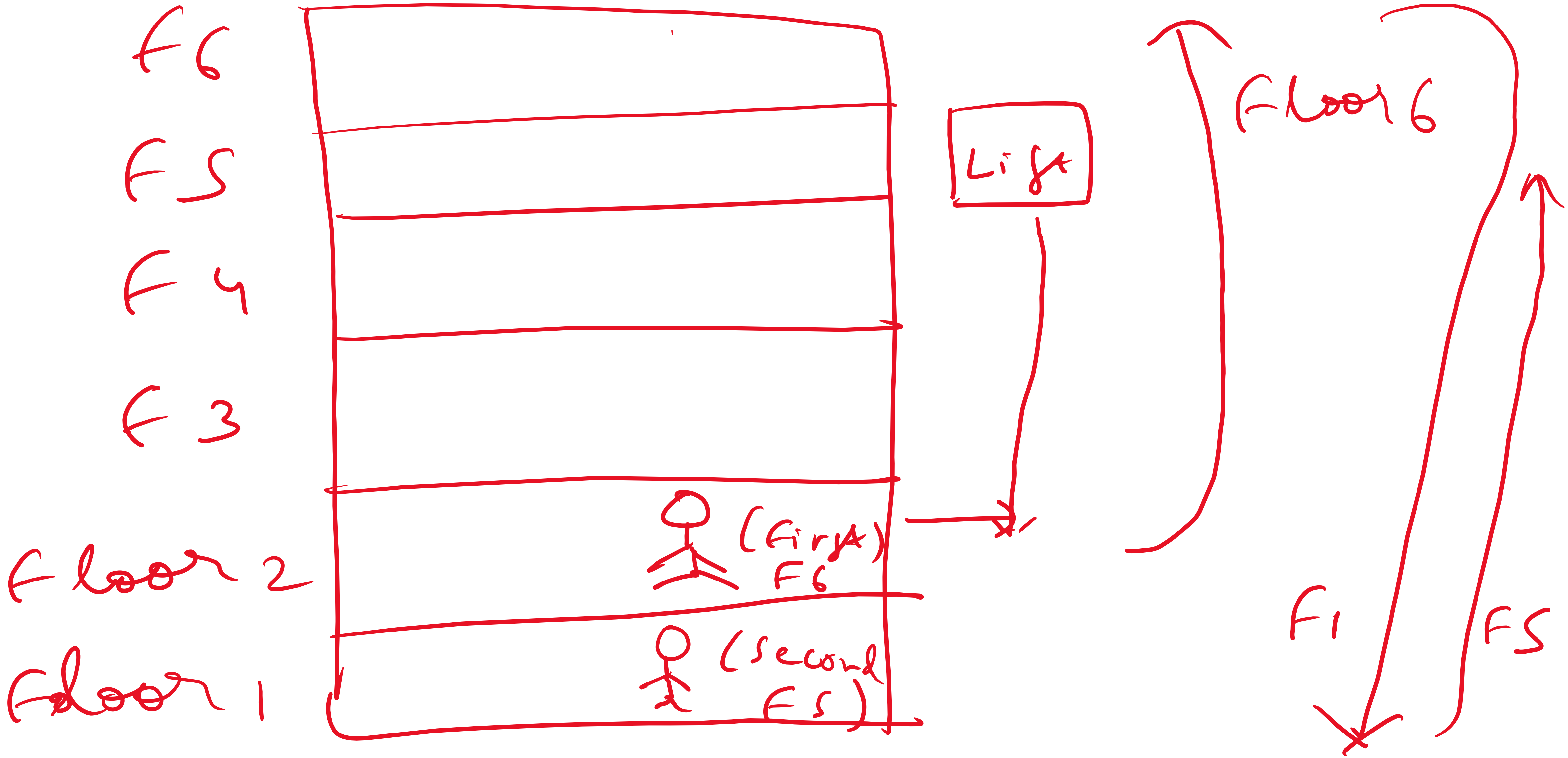
open, close


Building
Elevator System

(1) Different Dispatching Algorithm :-

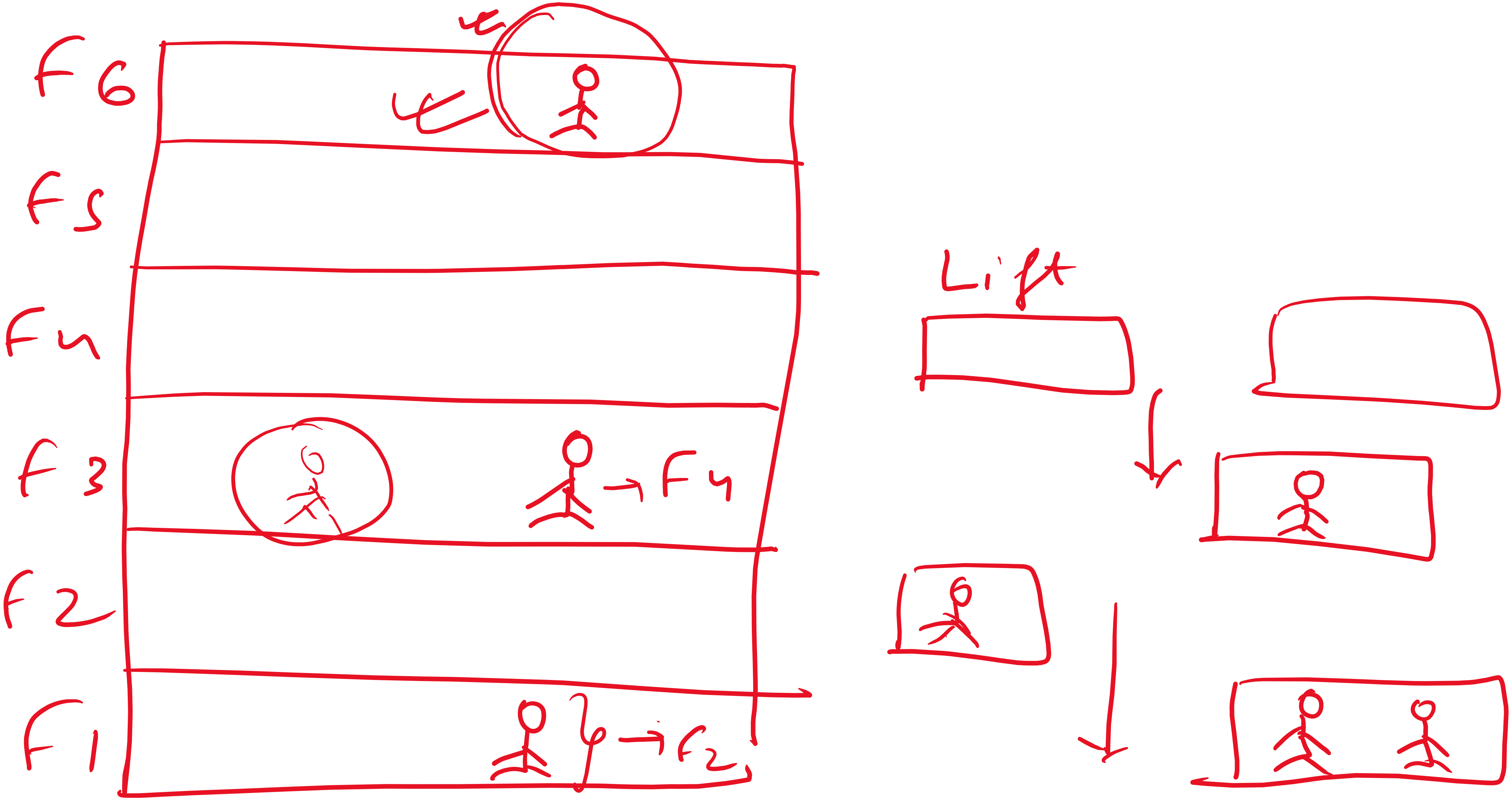
1. First Come First Serve (FCFS) :-

The passenger who comes first gets the elevator car and reaches the destination.


F6
F5
F4
F3
Floor 2
Floor 1
Lift
(First) F6
(Second F5)
Floor 6
F1
F5

2. Shortest Seek Time First (SSTF):-

The passenger who is closest to the elevator car would get the elevator car.


F6
F5
F4
F3
F2
F1
→ F4
→ F2
Lift

3. <u>SCAN</u> :- The Elevator car starts from one end of the building and moves towards the other end, servicing request that comes in between.

4. <u>LOOK</u> :- Here the elevator car stops when there is no request in front of them. It will move on the basis of the request. the advantage is that elevator car does not always go till the end of the building but can change it's direction in between.